Check for updates

ASME Journal of Biomechanical Engineering
Online journal at:
https://asmedigitalcollection.asme.org/biomechanical

Maryam Nazem
Center for Orthopaedic Biomechanics,
Department of Mechanical Engineering,
University of Denver,
Denver, CO 80210

Thor E. Andreassen
Center for Orthopaedic Biomechanics,
Department of Mechanical Engineering,
University of Denver,
Denver, CO 80210

Nancy Kim
Auckland Bioengineering Institute,
University of Auckland,
Auckland 1010, New Zealand;
Department of Engineering Science,
University of Auckland,
Auckland 1010, New Zealand

Kate Moyle
Auckland Bioengineering Institute,
University of Auckland,
Auckland 1010, New Zealand;
Department of Engineering Science,
University of Auckland,
Auckland 1010, New Zealand

Thor F. Besier
Auckland Bioengineering Institute,
University of Auckland,
Auckland 1010, New Zealand;
Department of Engineering Science,
University of Auckland,
Auckland 1010, New Zealand

Jason P. Halloran
Applied Sciences Laboratory,
Institute for Shock Physics,
Washington State University,
Spokane, WA 99202

Carl W. Imhauser
Department of Biomechanics,
Hospital for Special Surgery,
New York, NY 10021

Snehal Chokhandre
Department of Biomedical Engineering,
Lerner Research Institute,
Cleveland Clinic,
Cleveland, OH 44195

Deciphering the "Art" in Modeling and Simulation of the Knee Joint: Model Benchmarking

Given the strong ties to data sharing and the responsible use of resources, reproducibility of modeling and simulation practice is of paramount importance in science. Computational models in orthopedics provide insight into healthy and injured joint mechanics and can inform clinical decision-making. The KneeHub project investigated the influence of modelers' decisions and thus their "art" in simulation and modeling; five teams developed and calibrated knee models using the same experimental data. Model benchmarking evaluated the predictive ability of the models under loading scenarios that were not considered in the development and calibration process. The objective of this study was to evaluate the accuracy of predictions of knee-specific joint biomechanics for benchmark scenarios of simulating a resected anterior cruciate ligament (ACL) using models of one knee and a combined pivot shift loading using models of another knee. The models predicted the major trends in kinematics and kinetics; however, differences were observed in comparison to experimental data and between teams. Model-to-experiment root-mean-square (RMS) errors were up to 6.6 ± 2.4 mm in anterior–posterior (AP) translation, 13.5 ± 12.9 deg in internal–external (IE) rotation, and 5.3 ± 3.4 deg in varus–valgus (VV) rotation; errors were largest in internal–external rotation, and standard deviations reflected differences between teams. While calibrated models were tuned to a similar set of conditions (albeit with different decisions), the optimized stiffness and reference length/strain of ligament structures may not fully reproduce the contributions of these structures to joint kinematics that were measured experimentally in the benchmark scenarios. As researchers often extend models beyond the conditions used to calibrate them, quantifying model accuracy and limitations with benchmarking represents a crucial step toward reproducibility and can help establish best practices for credible modeling in our community.
[DOI: 10.1115/1.4070823]

Keywords: computational modeling, reproducibility, knee biomechanics, finite element analysis, joint mechanics, tissue mechanics, ligament, verification and validation

[1]Corresponding author.
Manuscript received July 18, 2025; final manuscript received January 2, 2026; published online February 5, 2026. Assoc. Editor: Matty J. Major.

Copyright © 2026 by ASME; reuse license CC-BY 4.0

Downloaded from http://asmedigitalcollection.asme.org/biomechanical/article-pdf/148/5/051004/7584244/bio-25-1200.pdf by University of Denver user on 08 March 2026

Marco T. Y. Schneider
Auckland Bioengineering Institute,
University of Auckland,
Auckland 1010, New Zealand;
Department of Engineering Science,
University of Auckland,
Auckland 1010, New Zealand

Shady Elmasry
Department of Biomechanics,
Hospital for Special Surgery,
New York, NY 10021

William Zaylor
Department of Biomedical Engineering,
Lerner Research Institute,
Cleveland Clinic,
Cleveland, OH 44195

Kevin B. Shelburne
Center for Orthopaedic Biomechanics,
Department of Mechanical Engineering,
University of Denver,
Denver, CO 80210

Ahmet Erdemir
Department of Biomedical Engineering,
Lerner Research Institute,
Cleveland Clinic,
Cleveland, OH 44195;
Computational Biomodeling (CoBi) Core,
Lerner Research Institute,
Cleveland Clinic,
Cleveland, OH 44195

Peter J. Laz[1]
Center for Orthopaedic Biomechanics,
Department of Mechanical Engineering,
University of Denver,
2155 East Wesley Avenue,
Denver, CO 80210
e-mail: Peter.Laz@du.edu

1 Introduction

Reproducibility is fundamental to the integrity of scientific research, particularly as data and model sharing become more prevalent [1–4]. Clear and transparent methods, open-source datasets, and comprehensive documentation enable the research community to evaluate findings and build upon existing work [5–8]. Recent efforts promoting reproducibility are targeted at improving the quality and strengthening the credibility of scientific conclusions [9], especially in high-stakes applications such as clinical decision-making.

Funding agencies emphasize responsible resource allocation by prioritizing reproducible studies, ensuring that research efforts yield reliable outcomes [4]. Similarly, regulatory bodies such as the U.S. Food and Drug Administration rely on transparent and verifiable methods to certify products safely and effectively. The strong connection between reproducibility, data integrity, and responsible resource use enables corroboration of findings by multiple researchers, helps to establish best practices, and, regarding physics-based modeling and simulation, is essential for developing consistent and verifiable workflows that foster trust among researchers and clinicians [10–12].

Over the last decade, standards and guidelines for credible practices in physics-based modeling and simulation have been developed by communities [1,5,6,13] and professional associations, such as the American Society of Mechanical Engineers [3], using a structured approach encompassing context of use, risk assessment, and evaluation of the appropriateness of model and data sources. Test cases and experimental data are integrated into modeling workflows to perform validation and sensitivity studies to understand the impact of uncertainty on model predictions [14–19]. The structured American Society of Mechanical Engineers Verification and Validation (V&V40) framework has been applied to support model credibility in assessments of hip fracture risk [20] and the development of medical devices [21].

With a focus on improving reproducibility in the biomechanics community, the KneeHub project was established to investigate the influence of modelers' decisions and the inherent subjectivity or art in modeling and simulation [22]. Specifically, multiple teams performed modeling and simulation of the same knee, based on the same experimental data but using their preferred workflow, which included team-specific decisions related to the representation of knee structures, coordinate systems, selection of tools, etc. In this

Downloaded from http://asmedigitalcollection.asme.org/biomechanical/article-pdf/148/5/051004/7584244/bio-25-1200.pdf by University of Denver user on 08 March 2026

way, the KneeHub project assessed broadly the concept of reproducibility by comparing predictions between researchers in the community, which may be considered different from repeatability or replicability, where errors are assessed using the same methods and tools, and often by the same researcher or within the same laboratory [1,4,10].

The KneeHub project consists of four phases: model development, model calibration, model benchmarking, and model reuse (Fig. 1). The anticipated contexts of use for the project are intended to demonstrate the use of knee joint models in biomechanically and clinically relevant scenarios. These scenarios include passive flexion (a foundational behavior of knee movements guided by articulating surfaces and ligamentous constraints); pivot shift exam (used clinically to diagnose injury of the anterior cruciate ligament (ACL)) [23,24]; quiet standing (such as during standing X-ray imaging); and a sit-to-stand simulation (as an activity of daily living) [22]. Publicly available imaging and experimental joint mechanics data from cadaveric knee specimens were used for model development and calibration, namely, Open Knee(s) [25,26] and Natural Knee [27,28]. In the model development phase, five independent teams constructed knee models from the available data using their preferred workflow [29,30]. In model calibration, each team selected kinetic and kinematic conditions from experimental joint mechanics data and the model parameters that were used for calibration of ligamentous material properties [31]. During model development and calibration, considerable differences emerged between teams regarding key decisions, e.g., representing passive restraints such as ligaments with nonlinear springs or as three-dimensional structures, selecting unique subsets of the available experimental data, defining calibration variables, and choosing optimization methods. While all teams optimized ligament stiffness and/or reference lengths/strains in ways they thought would most closely reproduce the experimental data, teams differed in the target data selected for calibration. For instance, some teams used all available laxity data at a single flexion angle, while others used the extremes of the laxity at multiple flexion angles, and one team used a subset of the available data [31]. Throughout the model development and model calibration phases, teams were free to choose their own strategies, highlighting differences in how teams chose to model the mechanical behavior of representative biomechanical systems.

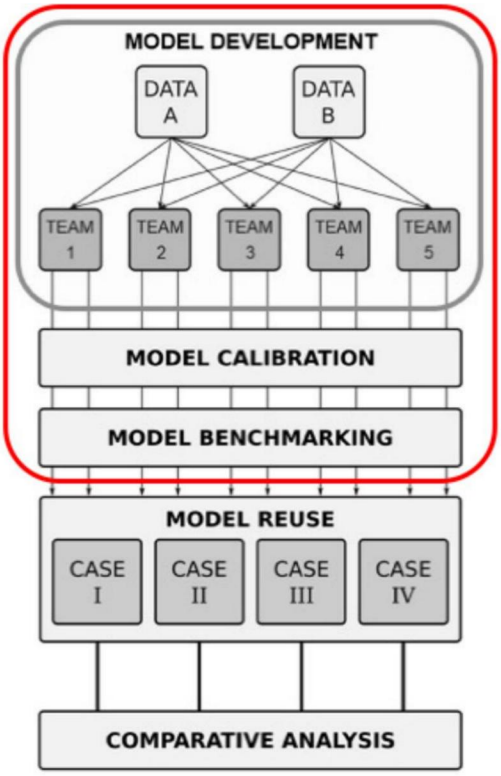

Fig. 1 Phases of the KneeHub project. Adapted from Erdemir et al. [22]. Boxed region shows phases completed with the current study presenting the findings for model benchmarking. DATA A refers to the Natural Knee specimen DU02, and DATA B refers to the Open Knee(s) specimen oks003.

The current study focuses on model benchmarking to evaluate how well the models reproduced knee joint mechanics in response to loading conditions and knee states beyond those used in the calibration phase. The benchmarking conditions were designed to establish relevance to clinical applications while leveraging available joint mechanics data from physical experiments. Accordingly, the objective of the current study was to evaluate the accuracy of knee-specific joint biomechanics predictions for benchmark scenarios based on models and data for two independent knees: one knee each from the Open Knee(s) (oks003) and Natural Knee (DU02) datasets. The benchmark scenarios included: removing the ACL, a key passive stabilizer of the knee, and assessing laxity behavior and passive flexion in the DU02 knee; also, applying a challenging combined, i.e., multidegree-of-freedom, loading condition, similar to the pivot shift exam used for diagnosing ACL injuries, in the oks003 knee. Thus, benchmarking evaluated whether the relative biomechanical contributions of the structures of the knee were adequately represented to describe knee biomechanics. While differences from prior phases carried forward, model benchmarking focused on quantifying the accuracy of model estimates in conditions that were not used for calibration to evaluate model robustness, identifying factors (decisions) that may have affected these estimates, and documenting limitations. Altogether, this work will establish the continuum of best practices to support reproducible and credible computational modeling.

2 Materials and Methods

2.1 Model Development and Model Calibration Phases. The model development and model calibration phases of the KneeHub project have been previously documented in Refs. [29] and [31]. Briefly, each of the five participating teams developed and calibrated two knee joint models using publicly available experimental datasets from Open Knee(s) [26,32] and Natural Knee Data [28]. Specimen oks003 (Open Knee) was a left knee of a 25-year-old female donor (height: 1.73 m; mass: 68 kg; BMI: 22.8). Specimen DU02 (Natural Knee Data) was a right knee of a 44-year-old male donor (height: 1.83 m; mass: 70.31 kg; BMI: 21.02). The experimental dataset included magnetic resonance imaging (both specimens), computed tomography (DU02 only), and mechanical testing data of tibiofemoral joint kinematics and kinetics [33]. These data were passive flexion (0–120 deg) and joint laxity testing for anterior–posterior (AP) translation, internal–external (IE) rotation, and varus–valgus (VV) rotation at multiple flexion angles [27]. The joint laxity measurements of the intact knee for AP translation, IE rotation, and VV rotation were used in the model calibration phase [31].

Each of the teams—Auckland Bioengineering Institute (ABI), Cleveland Clinic (CC), University of Denver (DU), Hospital for Special Surgery (HSS), and Washington State University (WSU)—applied its preferred workflow to develop and calibrate specimen-specific models of each knee (Fig. 2). The relevant anatomy was segmented from the imaging data, and anatomical coordinate systems were assigned [29]. Models included representations of bones, cartilage, and soft tissue structures. Teams used a variety of representations; models included detailed representations of the proximal tibia and distal femur. The bony structures were modeled as rigid bodies by all teams as the focus of the study was on joint-level kinematics. Most teams represented cartilage contact as rigid with a pressure–overclosure relationship, except for one team (CC) that applied a nearly incompressible Neo-Hookean material model [29]. Models for all teams included the ACL, posterior cruciate ligament (PCL), medial collateral ligament, and lateral collateral

Downloaded from http://asmedigitalcollection.asme.org/biomechanical/article-pdf/148/5/051004/7584244/bio-25-1200.pdf by University of Denver user on 08 March 2026

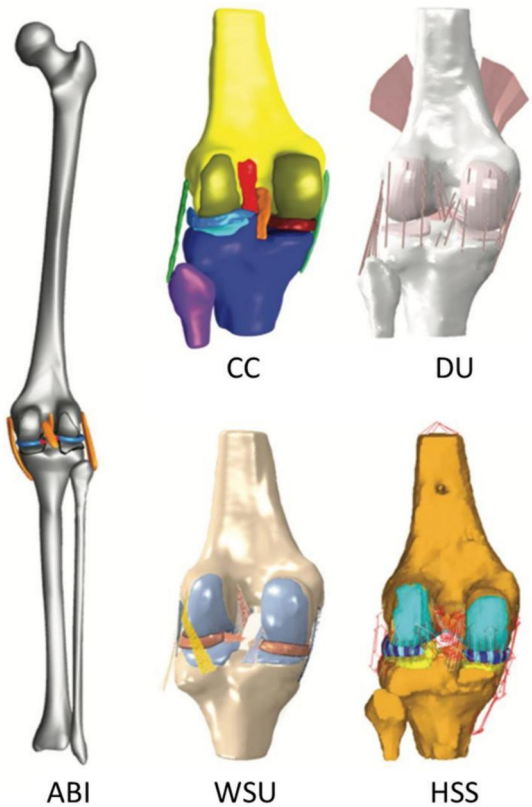

Fig. 2 Models of the oks003 knee developed by each team in the KneeHub project. Adapted from Rooks et al. [29]. Teams are abbreviated as: Auckland Bioengineering Institute (ABI), Cleveland Clinic (CC), University of Denver (DU), Hospital for Special Surgery (HSS), and Washington State University (WSU).

ligament, while most teams also included additional structures, e.g., popliteofibular ligament, oblique popliteal ligament, and posterior oblique ligament, among others [29]. Ligaments were represented as full 3D structures (ABI, CC) or as a series of nonlinear springs (DU, HSS, WSU) [29]; teams using nonlinear springs split structures into multiple bundles matching anatomical descriptions, e.g., two bundles for the ACL [34], two bundles for the PCL [35], and three or four bundles for the medial collateral ligament [36–38]. Ligament attachment site locations were identified using MR images and descriptions from the literature [37,39,40], but notably varied from team to team [29]. In model calibration, teams optimized the stiffness and/or slack lengths/reference strains of ligamentous structures to best match model predictions of joint-level kinematics with the experimental laxity behavior of the knee at their chosen calibration targets (knee flexion angles and applied external loads) [31].

In each modeling phase, teams prepared specification documents describing the intended modeling and simulation procedures. Protocol deviations were explicitly described and justified, and the resulting influence on the simulation outcomes was explained. Documents for the specification, protocol deviations, and models are publicly available on the SimTK project site [32].

2.2 Model Benchmarking Phase. Data earmarked for use in the model benchmarking phase are described below. These data were not used in the prior model development and model calibration phases. For the DU02 knee, experimental data following ACL resection were available for AP, IE, and VV laxity tests at two flexion angles (15 deg and 55 deg) and passive flexion from 0 deg to 120 deg. For the oks003 knee, joint mechanics measurements were available for the intact knee during a combined IE and VV loading condition at four flexion angles (0 deg, 30 deg, 60 deg, and 90 deg). The experimental data for all cases included kinetics and kinematics for all six degrees-of-freedom (DOF) for each activity. Kinematics were reported describing the relative motion of the tibia with the femur fixed. Selection of data for benchmarking accommodated data availability (from Open Knee(s) and Natural Knee Datasets) and relevance to anticipated contexts of use for the developed models in the model reuse phase [22].

Simulations were performed using the calibrated models from the prior phase. Models were initially positioned in full extension primarily corresponding to the imaged state (assumed zero loading state of the model). For simulations, loading and boundary conditions were initially applied to move the models to the full extension state of the experiments, i.e., the start of the experiments where the physical specimens were assumed to be at a zero load state. In the DU02 model, the ACL was removed, and all other structures were retained with their calibrated properties from the intact case. For the laxity test loading condition (DU02 knee), the knee was flexed to 15 deg or 55 deg, and loading of ± 80 N AP force, ± 8 N·m IE torque, or ± 10 N·m VV torque was applied per the experiment. As the bidirectional forces or moments were applied, e.g., anterior and posterior, it was common for the data to be generated with a pair of analyses for each direction. As the flexion angle was fixed, kinematics were reported for the remaining five DOF. Kinematics were reported at five points along the laxity curve, representing the neutral unloaded position of the experiment and the position with 50% and 100% of the magnitude of the force or moment applied in each direction (e.g., anterior and posterior).

For the passive flexion loading condition (DU02), the knee was flexed from 0 deg to 120 deg with no forces or moments applied. Passive flexion was previously evaluated for the intact knee in the model calibration phase; both the ACL-deficient and intact knee simulations during passive flexion were reported alongside the experimental data to consider the role of the ACL in joint kinematics.

For the combined IE and VV loading conditions (oks003), the intact knee was flexed to the prescribed flexion angle (0 deg, 30 deg, 60 deg, and 90 deg), and a combined loading of 5 N·m internal and 10 N·m valgus torques was applied to the tibia with the femur fixed to simulate a clinical pivot shift test. As the flexion angle was fixed, kinematics were predicted and compared to the experimental data for the other five DOF.

While applied forces and moments were available for all DOF in the experiment, teams made individual modeling decisions (in the model benchmarking phase) that influenced the predictions. For the DU02 laxity tests, two teams (ABI and HSS) applied idealized loading conditions, ramping the primary DOF and leaving the knee free in other DOF (e.g., ramping AP force and leaving IE and VV free for the AP test). Three teams (CC, DU, and WSU) applied the full set of kinetics to the model as measured experimentally.

2.3 Methods for Comparison. Various comparisons were made between the model predictions and experimental data. As teams performed simulations with their own coordinate system conventions, all results were converted to the coordinate system of the experiment. The DU02 experiment used absolute clinical Grood and Suntay joint coordinate system rotations and translations, while the oks003 experimental coordinate system used cylindrical Grood and Suntay joint rotations and translations relative to a neutral full-extension experimental pose [22,41].

Root-mean-square (RMS) errors were calculated for each DOF (AP, IE, and VV) for each loading condition between the model prediction and the experimentally measured value. Standard deviations were computed to describe the variation in RMS error among the teams. For some cases in the combined loading evaluation (oks003), model predictions for two teams had numerical convergence and instability issues; those results were not included in the RMS error calculation.

3 Results

In general, model predictions showed similar trends to the experiment in all simulations. However, differences with the experiment and between teams were observed, which are elaborated

Downloaded from http://asmedigitalcollection.asme.org/biomechanical/article-pdf/148/5/051004/7584244/bio-25-1200.pdf by University of Denver user on 08 March 2026

below. A summary of the RMS errors compared model predictions to the experimental data for all the benchmark scenarios (Table 1).

3.1 Laxity Test Comparisons for DU02 Knee. For the DU02 knee in the ACL-deficient condition, model predictions of the AP, IE, and VV laxity responses were compared with the experiment at two flexion angles, 15 deg and 55 deg (Fig. 3). In the loaded DOF, model-to-experiment RMS errors were 6.6±2.4 mm for AP translation in the AP laxity test, 8.8±4.2 deg for IE rotation in the IE laxity test, and 5.3±3.4 deg for VV rotation in the VV laxity test (Table 1). Standard deviations reflected differences between teams. Comparing predicted and experimental laxity responses, differences were classified as offset errors, describing the unloaded initial position with shift of the kinematic data from the intact knee models' reference state, or as range errors, representing differences in the bounds or extents of the kinematic data and linked to the relative stiffness of the knee's response.

The AP laxity predictions at both flexion angles did not capture the range observed in the experiment, and larger offset errors among the teams were observed at 55 deg flexion (Fig. 4). Model predictions were stiffer, particularly for 15 deg flexion, as observed by the smaller range of translation for the same magnitude of applied force (Fig. 3). Results for some DOFs had closer agreement with the experiment in one direction of applied loading than in the opposite direction. For the applied posterior tibial force at 55 deg flexion, the

Table 1 Summary table of RMS errors comparing model predictions against experiment for anterior–posterior (AP), internal–external (IE), and varus–valgus (VV) kinematics

Model	Test	AP (mm)	IE (deg)	VV (deg)
	AP laxity	**6.59 ± 2.39**	13.53 ± 12.85	3.74 ± 1.96
Natural Knee (DU02)	IE laxity	4.69 ± 2.98	**8.76 ± 4.18**	3.34 ± 1.70
No ACL	VV laxity	4.98 ± 3.37	8.81 ± 7.41	**5.31 ± 3.35**
	Passive flexion	6.12 ± 5.30	7.39 ± 5.83	3.97 ± 1.86
Open Knee (oks003)	Combined IE and VV loading	1.61 ± 0.78	**7.29 ± 3.77**	**1.71 ± 0.92**

Mean and standard deviation across teams are reported. Bold text refers to degrees-of-freedom that were actively loaded.

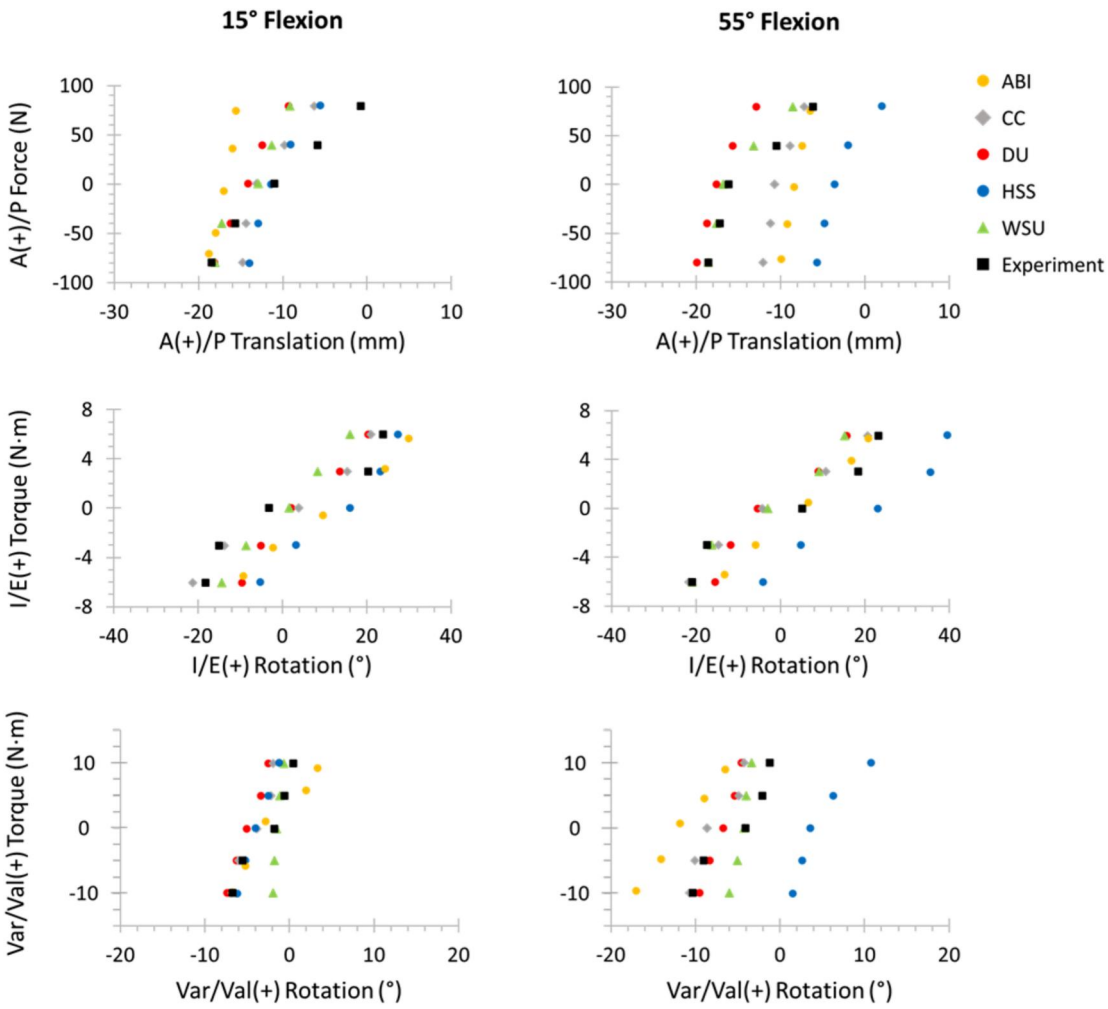

Fig. 3 Anterior–posterior (AP), internal–external (IE), and varus–valgus (Var/Val) force/torque versus translation/rotation laxity response for model benchmarking of the DU02 knee in the ACL-deficient condition at 15 deg (left) and 55 deg (right). Legend refers to teams: Auckland Bioengineering Institute (ABI), Cleveland Clinic (CC), University of Denver (DU), Hospital for Special Surgery (HSS), and Washington State University (WSU).

Downloaded from http://asmedigitalcollection.asme.org/biomechanical/article-pdf/148/5/051004/7584244/bio-25-1200.pdf by University of Denver user on 08 March 2026

relative behavior agreed closely, differing by $\leq$1.0 mm, with the experiment for all teams, while more variation among the teams and greater differences with the experiment (ranging from 1.8 to 8.1 mm) were observed for the applied anterior tibial force (Fig. 4). To isolate and visualize the differences in range and stiffness response, model data for each team were also shifted by subtracting the unloaded kinematic position so that the laxity response began at the origin (Fig. S1 available in the Supplemental Materials on the ASME Digital Collection).

In IE and VV laxity tests, differences between team predictions and the experiment were comprised of a combination of offset and range errors (Figs. 3 and 4). The offset error describing the neutral unloaded position was highly variable between teams; the predicted range of IE rotation, related to the stiffness response of the knee, agreed more closely with average range errors of 16%. In the experiment, VV ranges (7.2 deg and 9.1 deg at 15 deg and 55 deg flexion) were smaller than IE ranges (42.1 deg and 44.0 deg). Model predictions for VV were the only DOF in which some models predicted a greater range than the experiment of up to 2.9 deg.

The RMS errors were computed for the primary loaded DOF, but also considered secondary DOF, e.g., IE and VV kinematic errors were reported during the AP laxity test (Table 1). In the secondary DOFs, RMS errors were up to 5.0$\pm$3.4 mm in AP translation, 13.5$\pm$12.9 deg in IE rotation, and 3.7$\pm$2.0 deg in VV rotation. There were no clear trends of errors being smaller in primary loaded DOF than secondary DOFs. Across all laxity comparisons, the largest errors occurred in the IE DOF of up to 13.5 deg. A more detailed breakdown of errors by team and specific DOFs can be found in the Supplemental Materials.

3.2 Passive Flexion Comparison for DU02 Knee. For the passive flexion simulation (DU02), kinematic results were reported for the experiment and model predictions in the ACL intact (model calibration) and ACL-deficient (model benchmarking) conditions (Fig. 5). Over the flexion range of 0 deg to 120 deg, the experimental data showed little difference in the AP, IE, and VV kinematics between the intact and ACL-deficient conditions. Model predictions for most of the teams exhibited similar behavior with small differences between the intact and ACL-deficient conditions, particularly for AP. Differences between the intact and ACL-deficient results were observed for some teams and DOFs, especially in deeper flexion (e.g., AP for Team CC and IE and VV for Team

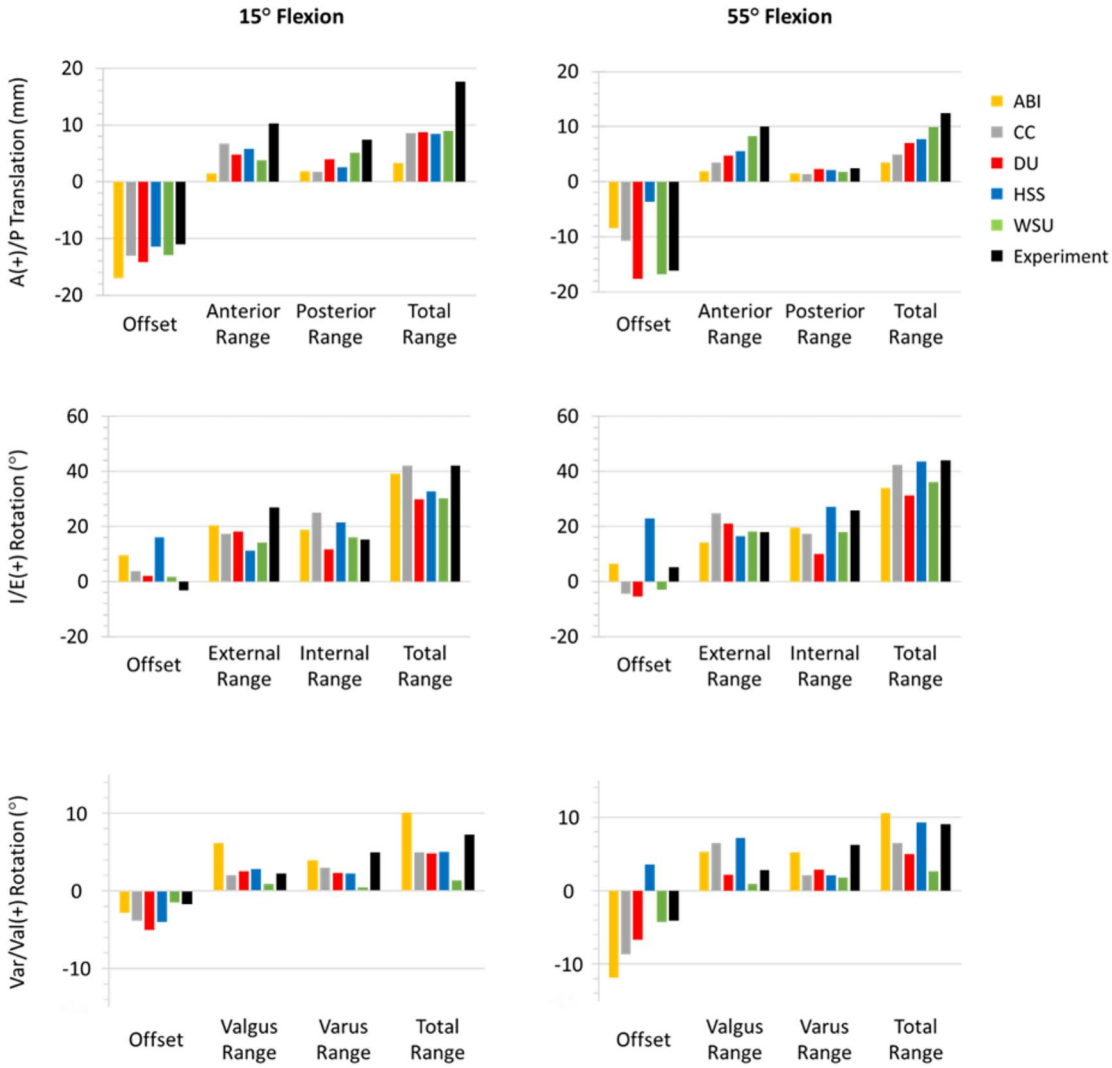

Fig. 4 Distribution of offset and range errors in the anterior–posterior (AP), internal–external (IE), and varus–valgus (Var/Val) laxity kinematics for the DU02 knee in the ACL-deficient condition at 15 deg (left) and 55 deg (right) flexion. Offset describes the initial unloaded position. Total range describes the kinematic bounds for each degree-of-freedom at 100% applied force or torque. One-dimensional ranges, e.g., anterior and posterior ranges, describe the amount of motion from the neutral position to the extent. The two one-dimensional ranges sum to match the total range.

Downloaded from http://asmedigitalcollection.asme.org/biomechanical/article-pdf/148/5/051004/7584244/bio-25-1200.pdf by University of Denver user on 08 March 2026

ABI). Model predictions for three teams did not reach the 100 deg flexion angle, and one model started from a 10 deg flexion angle. Errors for the intact models were lower than those without the ACL (Table 2); the IE degree had the highest RMS error and standard deviation.

3.3 Combined Loading Comparison for oks003 Knee. For the oks003 knee under combined IE and VV torques, the kinematic predictions generally followed the trends and magnitudes of the experimental data across the four flexion angles (Fig. 6). The model for one team (HSS) partially dislocated exhibiting excessive anterior translation of the lateral tibial compartment. This model was run with an explicit solver, and the representation of the attachment sites and slack lengths for the anterolateral ligament, ACL, and roots of the lateral meniscus may have contributed to nonphysiologic levels of internal tibial rotation. Similarly, the formulation for the PCL may have increased rollback of the lateral femoral condyle during passive flexion, also leading to nonphysiologic levels of internal tibial rotation. The model for another team (ABI) failed to converge at 90 deg flexion; this model used an implicit formulation with fully deformable elements, and thus some elements of the mesh may have had large deformations impacting convergence. Data were reported at ~75 deg flexion, where the model was able to converge. These data points were excluded in the calculation of errors. All predictions for all teams are in the Supplemental Materials (Fig. S2) on the ASME Digital Collection.

With applied IE and VV torques, the errors were smaller for AP (1.6±0.8 mm) and VV (1.7±0.9 deg) kinematics than for IE (7.3±3.8; Tables 1 and 3). The IE DOF had the largest error and largest standard deviation reflecting the variability in the team results. The magnitudes of the errors should also be considered in the context of the amount of motion, with the knee moving more in IE than VV rotation.

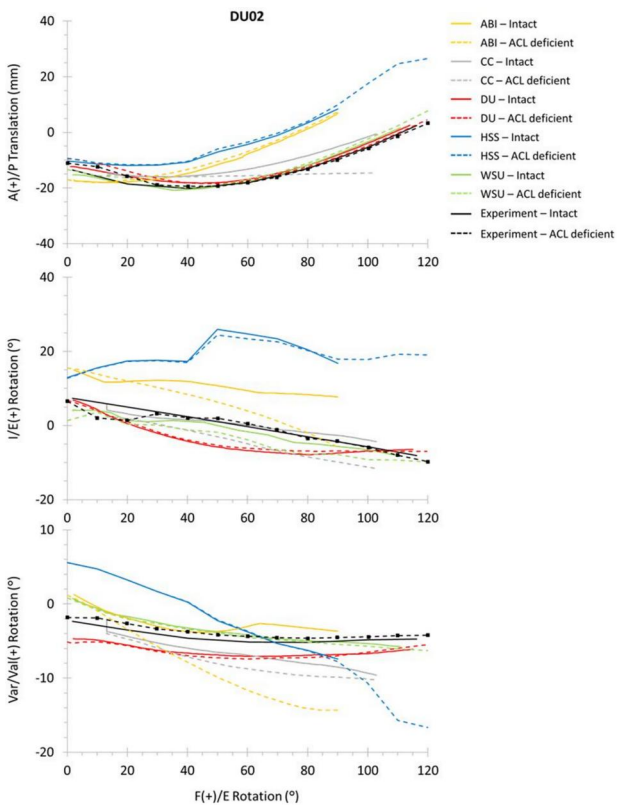

Fig. 5 Passive flexion kinematics for intact (model calibration) and ACL-deficient (model benchmarking) conditions for the DU02 knee. Anterior–posterior (AP), internal–external (IE), and varus–valgus (Var/Val) kinematics plotted as a function of flexion. Legend refers to teams: Auckland Bioengineering Institute (ABI), Cleveland Clinic (CC), University of Denver (DU), Hospital for Special Surgery (HSS), and Washington State University (WSU).

4 Discussion

By documenting modeling workflows and decision-making and sharing models and results, the goals of the KneeHub project are to enhance model reproducibility and to support the broader adoption of computational knee biomechanics in research and clinical applications. Prior publications have documented the overall strategy of the KneeHub project [22], model development [29], and model calibration [31] phases. Further, the KneeHub website documents all activities related to the comprehensive project [32], including experimental data sources, documentation, and models for each team. Notably, each team produced detailed protocols for each phase and protocol deviations, if any changes were made during implementation. Promoting reproducibility and credible modeling, this level of documentation provides a clear reference for fellow modelers with regard to model development practice, calibration and validation processes, and implications of modeling decisions.

As researchers often extend models beyond the conditions used to calibrate them, quantifying model accuracy and limitations with benchmarking represents an important step in model reproducibility. Models tend to do better when operating in the range of data used to tune them, and benchmarking intentionally pushes models outside of the range of data used to calibrate them. Across the benchmark scenarios, most models were able to reproduce general trends in the experimental behavior; however, model prediction errors were 6.6±2.4 mm for AP, 13.5±12.9 deg in IE, and 5.3±3.4 deg in VV (Table 1). Given the complexity of the joint and the interaction of multiple complex structures, this level of accuracy can be considered reasonable, but it is not clear whether it is sufficient for the desired context of use of applying models to diagnose mechanical consequences of pathology and inform clinical treatment. For context, Engebretsen et al. [42] reported differences in anterolateral subluxation of up to 5.8±1.6 mm between intact and ACL-resected knees in cadaveric experiments under pivot shift loading conditions. Further, Harris et al. [27] reported experimental AP range of motion experiments of 9.0±3.0 mm (intact) and 18.3±10.3 mm (resected ACL) at 15 deg flexion and 7.3±2.1 mm (intact) and 22.8±5.5 mm (resected ACL) at 60 deg flexion. The AP laxity experiment for DU02 in an ACL-deficient condition exhibited a range of motion of 17.7 mm at 15 deg flexion and 12.4 mm at 55 deg flexion. The errors between model prediction and experiment of 6.6±2.4 mm for AP translation in this study had a similar magnitude to the differences between the intact and resected ACL conditions in Refs. [27] and [42].

In the DU02 knee, benchmarking evaluations were focused on an ACL-deficient condition, noting that the knee models were calibrated using data with the ACL intact. With the ACL removed, the laxity behavior was generally stiffer with models under-predicting the amount of translation observed in the experiment. The ACL restricts anterior tibial translation, and the experiment showed greater anterior translation than model predictions for an applied anterior force (Fig. 3). Differences between model and experiment were comprised of both offset and range errors (Fig. 4) and may be influenced by factors, including offsets in kinematic position of the knee in the unloaded state between the intact and ACL-deficient states and differences in the relative contributions of the ligaments. In passive flexion, the experiment and model predictions did not show substantial differences between the intact and ACL-deficient conditions. Most of the models successfully captured the kinematic trends (Fig. 5) with RMS errors between model and experiment slightly smaller for the intact condition than the ACL-deficient condition (Table 2). It was observed that teams using 3D representations (ABI, CC) showed greater differences between the intact and ACL-deficient conditions, particularly in deeper flexion, compared to the teams using multiple spring representations. During passive flexion, the knee was relatively

Downloaded from http://asmedigitalcollection.asme.org/biomechanical/article-pdf/148/5/051004/7584244/bio-25-1200.pdf by University of Denver user on 08 March 2026

Table 2 Root-mean-square (RMS) error comparison for the DU02 knee under passive flexion loading

	With ACL (calibration)			Without ACL (benchmarking)		
Team	AP (mm)	IE (deg)	VV (deg)	AP (mm)	IE (deg)	VV (deg)
ABI	3.51	9.70	2.19	10.77	5.99	6.38
CC	3.07	1.43	2.43	3.62	4.80	4.29
DU	0.92	4.56	2.17	1.52	4.31	2.99
HSS	10.99	18.06	5.01	12.85	17.74	4.78
WSU	1.53	2.27	1.83	1.86	4.11	1.45
Mean ± standard deviation	4.00±4.05	7.20±6.87	2.73±1.30	6.12±5.30	7.39±5.83	3.97±1.86

RMS errors reported in anterior–posterior (AP), internal–external (IE), and varus–valgus (VV) kinematic degrees-of-freedom for models in the intact and ACL-deficient conditions. The mean value and standard deviation are reported to compare team results. Teams are abbreviated as: Auckland Bioengineering Institute (ABI), Cleveland Clinic (CC), University of Denver (DU), Hospital for Special Surgery (HSS), and Washington State University (WSU).

unloaded, and the constraints of the articular surfaces may have played a larger role in the behavior than contributions from the soft tissue constraints. The formulation of the PCL, a key driver of posterior femoral translation (e.g., femoral rollback with increased flexion), is also critical in passive flexion [43,44]. Key parameters that may affect rollback regarding the PCL are its insertion on the femur and pretension on the ligament fibers, particularly at the anterolateral portion of this ligament [45].

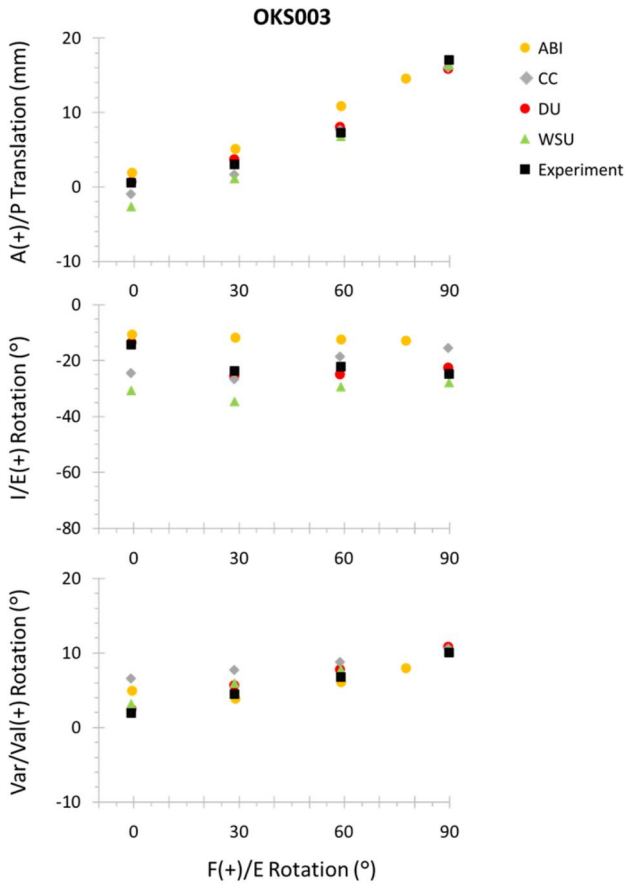

Fig. 6 Anterior–posterior (AP), internal–external (IE), and varus–valgus (Var/Val) kinematic comparison for the oks003 knee under combined IE and VV loading at four flexion angles. Results are shown for the experiment and four teams, as the model for one team partially dislocated. One team (ABI) failed to converge at 90 deg flexion; a converged result at 75 deg flexion is shown. Legend refers to teams: Auckland Bioengineering Institute (ABI), Cleveland Clinic (CC), University of Denver (DU), Hospital for Special Surgery (HSS), and Washington State University (WSU).

For knee oks003, the benchmarking scenario applied a challenging loading condition simulating an applied internal rotation/torque and valgus torque of a pivot shift test [42,46] over a range of flexion angles from 10 deg to 90 deg. In model calibration, the soft tissue constraint was tuned to experimental data that involved single-DOF applied loading (i.e., laxity tests). While the errors in AP and VV were small, the mean error of 7.3 deg for IE rotation suggests that the soft tissue constraint did not fully replicate that of the physical knee experiment; model predictions were more variable, consistently over- or underpredicting the experimental behavior for the multiple flexion angles. While the errors for most teams' predictions were smallest for this activity (Table 1), the combined loading resulted in partial dislocation of the lateral tibial compartment for one model and caused numerical instability in another. These issues with model stability and convergence are an important consideration in reproducibility. For one team (HSS), decreased slack length and more anisometric placement of the anterolateral fibers of the PCL may have contributed to increased rollback and excessive internal tibial rotation, resulting in partial anterior dislocation of the lateral tibial compartment. Representation of the attachment sites and increased slack lengths of the anterolateral ligament, ACL, or roots of the lateral meniscus may also have contributed to nonphysiologic levels of internal tibial rotation. In the implicit formulation used by another team (ABI), complex models with deformable elements can fail to converge for a range of possible reasons, including contact nonlinearities, material and geometric nonlinearities, mesh and element formulation, solver and numerical parameters, and boundary conditions and constraints.

The accuracy of predictions in the model benchmarking phase was strongly tied to the modeling decisions and procedures from the model development and model calibration phases and has implications for model reuse and broader application. A range of representations for the ligaments was selected by the teams; two

Table 3 RMS errors for all teams in anterior–posterior (AP), internal–external (IE), and varus–valgus (VV) degrees-of-freedom for combined loading (oks003)

Team	AP (mm)	IE (deg)	VV (deg)
ABI[a]	2.56	9.26	1.83
CC	1.17	7.25	2.97
DU	0.81	2.04	0.95
WSU	1.90	10.63	1.09
Mean ± standard deviation	1.61 ± 0.78	7.29 ± 3.77	1.71 ± 0.92

Mean and standard deviation values describe differences between the teams in each degree-of-freedom. Teams are abbreviated as: Auckland Bioengineering Institute (ABI), Cleveland Clinic (CC), University of Denver (DU), Hospital for Special Surgery (HSS), and Washington State University (WSU). Models results for HSS were not included because of instability.
[a]Results at the 90 deg flexion angle were excluded for team ABI because of a convergence issue.

Downloaded from http://asmedigitalcollection.asme.org/biomechanical/article-pdf/148/5/051004/7584244/bio-25-1200.pdf by University of Denver user on 08 March 2026

teams used 3D representations with transversely isotropic hyperelastic behavior, while three teams used a series of nonlinear tension-only springs and represented ligaments with multiple bundles. Many studies have investigated the function and modeling of ligament structures [37,38,47,48], including discussion of bundle and fiber orientation and guidance on how to best model the native anatomy. Attachment site locations were derived from imaging and anatomical descriptions in the literature [35,37,49–52]. As noted in the model calibration phase, several teams selected properties for stiffness or modulus of the ligament structures from the literature and tuned slack length/reference strain in calibration, while one team optimized both stiffness and slack length/reference strain. Most teams applied optimization strategies using local and gradient-based algorithms with one team implementing a global simulated annealing approach. Teams using 3D ligament representations strategically selected experimental laxity data at one flexion angle and in a specific DOF to calibrate the material properties and/or reference strains for a specific structure. Teams using spring representations more commonly used kinematic/kinetic points representing the extents of the laxity response at multiple flexion angles and optimized ligament stiffness and/or reference strain for multiple structures to replicate the laxity behavior during a single-DOF experiment. Some teams considered other unloaded DOF (e.g., IE during an AP laxity test) in the calibration process. The influence of ligament representation, attachment sites, properties, and slack length/reference strain on the predicted knee response has been similarly observed in the literature [53–56].

In the model calibration phase, a range of values for the reference strains of the primary knee ligaments were found between teams and across the models for knees DU02 and oks003. While calibration of the properties and reference strain can closely reproduce an experiment in one DOF, it may not be sufficient to capture the behavior for other scenarios (e.g., removal of a structure, combined loading). In fact, the tuned parameter implicitly compensates for idealized or simplified conditions (or lack of representation of another structure). As loading conditions become more complex, these calibrated parameters may not reflect the accurate physiological behaviors, meaning that load sharing and structural compensation may differ. For instance, the one-directional behavior of a structure used in the calibration phase may not capture the nonlinear behavior that emerges under multidirectional strain and deformation during combined loading. Moreover, when carrying out an optimization-based calibration, there can be multiple solutions that replicate the joint-level behavior, but these solutions may not accurately reproduce the individual contributions of the various structures. For example, Theodorakos and Andersen [57] showed that optimization of ligament parameters was sensitive to the initial guesses used and can provide similar kinematics results for the same loading with very different ligament material properties and correspondingly different ligament force predictions. The differences in representation and calibration strategy described are part of the modeler's art and have implications and downstream impact on the accuracy and usability of models.

One of the modeling decisions in the model benchmarking phase was to apply the full set of 6DOF forces and moments from the experiment or an idealized representation of loading in the primary DOF with other DOFs unconstrained. Using idealized loading can affect model prediction accuracy due to differences between the loads applied in the experiment and those applied in the model. There were no clear trends observed in Figs. 1 and 2 or the associated errors between the teams that applied idealized loading (ABI, HSS) and those that applied the full 6DOF kinematics (CC, DU, WSU). This type of simplification could affect the evaluation by introducing errors that stem from assumptions regarding model inputs, not from the formulation of the model. Beyond loading and boundary conditions, variations in the way modelers represent anatomical structures also influenced the results; these decisions included the selection of ligament representation (2D or 3D), geometry, attachment sites, properties, and calibration strategy. All decisions made from model development to model benchmarking can have a downstream effect. Thus, a limitation of our study design is the difficulty in drawing a conclusion on the biomechanical impact of any single modeling decision as only post-hoc analyses have been performed on the simulation results of this study.

A further limitation is that statistical evaluations of significance were not carried out because of the small sample size with model predictions available for five teams. The benchmarking comparisons were also directly between model predictions and experiments for specific specimens. In this way, the study did not consider the impact of intersubject variability, and the results and relative findings for one specimen may not apply to other knees.

In model benchmarking, differences between model and experimental results are likely attributed to differences in the loading of structures and their relative contributions. While the modeled knees were calibrated with specimen-specific data, the availability of data is often limited in biomechanics as properties and load distributions are difficult to measure in tissue. Benchmarking performance may also be impacted by varied experimental setups and approaches that may limit the accuracy and relevance of benchmarking data and thus may introduce different implications on a model's suitability for its desired context of use. It is noted that the current modeling effort represents a data-rich scenario, and this level of data may not be available in other studies, in particular for clinical applications. Still, recent work [19] has validated subject-specific knee models that are calibrated using data from in vivo knee laxity devices. indicating that data-poor environments may have less impact on model accuracy than the differences in how these models are developed and calibrated.

Our most important conclusion is that the lack of a standardized or consensus approach to knee modeling and the resulting differences in modeling approach and decision-making influenced the accuracy and variability in the model benchmark predictions, namely, the ability of models to predict beyond the conditions used to calibrate them. By documenting and sharing models, results, and decision-making processes, the KneeHub project contributes to the broader effort of improving model reproducibility and credible modeling and simulation practices. The findings underscore the continued need for engagement of the community and consensus-based workflows to help establish best practices and support the broader adoption of computational knee biomechanics in research and clinical applications.

Author Contribution Statement

A.E., T.B., C.I., J.H., P.L., and K.S. designed the study. M.N. and P.L. outlined the document, wrote the first full draft, organized the data and results, and acquired input from coauthors. All co-authors performed modeling, added content in relevant sections, reviewed and edited the draft, and confirmed their agreement with the contents of the document.

Acknowledgment

The authors gratefully acknowledge Nynke Rooks, Donald Hume, Neda Nohouji, and Ariel Schwartz for their valuable contributions in the model development and model calibration phases, which were fundamental to the KneeHub project.

Contributions of the authors were primarily supported by the National Institute of Biomedical Imaging and Bioengineering, National Institutes of Health (NIH) (Grant No. R01EB024573). Efforts by A.E. were also supported in part by NIH R01GM104139, and K.S. and T.A. by NIH R01EB015497. The views, opinions, and/or findings contained in this document are those of the authors and do not necessarily reflect the views of the funding agencies.

Funding Data

- National Institute of Biomedical Imaging and Bioengineering, National Institutes of Health (NIH) (Grant No. R01EB024573; Funder ID: 10.13039/100000070).

Downloaded from http://asmedigitalcollection.asme.org/biomechanical/article-pdf/148/5/051004/7584244/bio-25-1200.pdf by University of Denver user on 08 March 2026

- National Institute of General Medical Sciences, National Institutes of Health (NIH) (Grant No. R01GM104139; Funder ID: 10.13039/100000057).
- National Institute of Biomedical Imaging and Bioengineering, National Institutes of Health (NIH) (Grant No. R01EB015497; Funder ID: 10.13039/100000070).

Conflict of Interest

The authors do not report any conflicts of interest with the content of this article.

Data Availability Statement

The data and information that support the findings of this article are freely available at: https://simtk.org/projects/kneehub.

References

[1] Goodman, S. N., Fanelli, D., and Ioannidis, J. P. A., 2016, "What Does Research Reproducibility Mean?," Sci. Transl. Med., **8**(341), p. 341ps12.
[2] Baker, M., 2016, "1,500 Scientists Lift the Lid on Reproducibility," Nature, **533**(7604), pp. 452–454.
[3] American Society of Mechanical Engineers, 2018, "VV40—Assessing Credibility of Computational Modeling Through Verification and Validation Application to Medical Devices," ASME, New York, accessed June 19, 2025, https://www.asme.org/codes-standards/find-codes-standards/assessing-credibility-of-computational-modeling-through-verification-and-validation-application-to-medical-devices
[4] National Institutes of Health, 2015, "Enhancing Reproducibility Through Rigor and Transparency," Grants & Funding, Bethesda, MD, accessed June 19, 2025, https://grants.nih.gov/policy-and-compliance/policy-topics/reproducibility
[5] Taylor, M., and Prendergast, P. J., 2015, "Four Decades of Finite Element Analysis of Orthopaedic Devices: Where Are We Now and What Are the Opportunities?," J. Biomech., **48**(5), pp. 767–778.
[6] Erdemir, A., Guess, T. M., Halloran, J., Tadepalli, S. C., and Morrison, T. M., 2012, "Considerations for Reporting Finite Element Analysis Studies in Biomechanics," J. Biomech., **45**(4), pp. 625–633.
[7] Shalhoub, S., Fitzwater, F., Eboch, W., Dickinson, M., Akhbari, B., Louie, E., Maletsky, L., et al., 2015, "In Vitro Experimental Testing of the Human Knee: A Concise Review," J. Knee Surg., **29**(2), pp. 138–148.
[8] Balci, O., 2012, "A Life Cycle for Modeling and Simulation," Simulation, **88**(7), pp. 870–883.
[9] Hubbard, D. W., and Carriquiry, A. L., 2019, "Quality Control for Scientific Research: Addressing Reproducibility, Responsiveness, and Relevance," Am. Stat., **73**(sup1), pp. 46–55.
[10] National Academies of Sciences, 2019, "Understanding Reproducibility and Replicability," Reproducibility and Replicability in Science, National Academies Press, Washington, DC, accessed July 08, 2025, https://www.ncbi.nlm.nih.gov/books/NBK547546/
[11] Bhakta, S., Xie, J., Pea, F., and Gillespie, S. H., 2025, "Data Transparency and Reproducibility in Health Research: Bridging the Gap for Early-Career Researchers," Front. Antibiot., **4**, p. 1562002.
[12] Belbasis, L., and Panagiotou, O. A., 2022, "Reproducibility of Prediction Models in Health Services Research," BMC Res. Notes, **15**(1), p. 204.
[13] Erdemir, A., Mulugeta, L., Ku, J. P., Drach, A., Horner, M., Morrison, T. M., Peng, G. C. Y., Vadigepalli, R., Lytton, W. W., and Myers, J. G., 2020, "Credible Practice of Modeling and Simulation in Healthcare: Ten Rules From a Multidisciplinary Perspective," J. Transl. Med., **18**(1), p. 369.
[14] Viceconti, M., Pappalardo, F., Rodriguez, B., Horner, M., Bischoff, J., and Musuamba Tshinanu, F., 2021, "In Silico Trials: Verification, Validation and Uncertainty Quantification of Predictive Models Used in the Regulatory Evaluation of Biomedical Products," Methods, **185**, pp. 120–127.
[15] Taylor, M., Bryan, R., and Galloway, F., 2013, "Accounting for Patient Variability in Finite Element Analysis of the Intact and Implanted Hip and Knee: A Review," Int. J. Numer. Methods Biomed. Eng., **29**(2), pp. 273–292.
[16] Ataei, A., Eggermont, F., Baars, M., van der Linden, Y., de Rooy, J., Verdonschot, N., and Tanck, E., 2021, "Evaluation of Inter- and Intra-Operator Reliability of Manual Segmentation of Femoral Metastatic Lesions," Int. J. Comput. Assist. Radiol. Surg., **16**(10), pp. 1841–1849.
[17] Henak, C. R., Anderson, A. E., and Weiss, J. A., 2013, "Subject-Specific Analysis of Joint Contact Mechanics: Application to the Study of Osteoarthritis and Surgical Planning," ASME J. Biomech. Eng., **135**(2), p. 021003.
[18] Liukkonen, M. K., Mononen, M. E., Tanska, P., Saarakkala, S., Nieminen, M. T., and Korhonen, R. K., 2017, "Application of a Semi-Automatic Cartilage Segmentation Method for Biomechanical Modeling of the Knee Joint," Comput. Methods Biomech. Biomed. Eng., **20**(13), pp. 1453–1463.
[19] Andreassen, T. E., Hume, D. R., Hamilton, L. D., Hegg, S. L., Higinbotham, S. E., and Shelburne, K. B., 2025, "Validating Subject-Specific Knee Models From In vivo Measurements," Front. Bioeng. Biotechnol., **13**, p. 1554836.
[20] Aldieri, A., Curreli, C., Szyszko, J. A., La Mattina, A. A., and Viceconti, M., 2023, "Credibility Assessment of Computational Models According to ASME V&V40: Application to the Bologna Biomechanical Computed Tomography Solution," Comput. Methods Programs Biomed., **240**, p. 107727.
[21] Nagaraja, S., Loughran, G., Baumann, A. P., Kartikeya, K., and Horner, M., 2024, "Establishing Finite Element Model Credibility of a Pedicle Screw System Under Compression-Bending: An End-to-End Example of the ASME V&V 40 Standard," Methods, **225**, pp. 74–88.
[22] Erdemir, A., Besier, T. F., Halloran, J. P., Imhauser, C. W., Laz, P. J., Morrison, T. M., and Shelburne, K. B., 2019, "Deciphering the &Art' in Modeling and Simulation of the Knee Joint: Overall Strategy," ASME J. Biomech. Eng., **141**(7), p. 071002.
[23] Vaudreuil, N. J., Rothrauff, B. B., de Sa, D., and Musahl, V., 2019, "The Pivot Shift: Current Experimental Methodology and Clinical Utility for Anterior Cruciate Ligament Rupture and Associated Injury," Curr. Rev. Musculoskelet. Med., **12**(1), pp. 41–49.
[24] Bull, A. M. J., and Amis, A. A., 1998, "The Pivot-Shift Phenomenon: A Clinical and Biomechanical Perspective," Knee, **5**(3), pp. 141–158.
[25] Chokhandre, S., Neumann, E. E., Nagle, T. F., Colbrunn, R. W., Flask, C. A., Colak, C., Halloran, J., and Erdemir, A., 2021, "Specimen Specific Imaging and Joint Mechanical Testing Data for Next Generation Virtual Knees," Data Brief, **35**, p. 106824.
[26] Erdemir, A., and Open Knee(s) Development Team, 2015, "Open Knee(s): Virtual Biomechanical Representations of the Knee Joint," Stanford, CA, accessed Jan. 12, 2025, https://simtk.org/projects/openknee
[27] Harris, M. D., Cyr, A. J., Ali, A. A., Fitzpatrick, C. K., Rullkoetter, P. J., Maletsky, L. P., and Shelburne, K. B., 2016, "A Combined Experimental and Computational Approach to Subject-Specific Analysis of Knee Joint Laxity," ASME J. Biomech. Eng., **138**(8), p. 081004.
[28] University of Denver, Center for Orthopaedic Biomechanics, 2016, "Natural Knee Data," University of Denver, Denver, CO, accessed Jan. 1, 2025, https://digitalcommons.du.edu/natural_ knee_data/
[29] Rooks, N. B., Schneider, M. T. Y., Erdemir, A., Halloran, J. P., Laz, P. J., Shelburne, K. B., Hume, D. R., et al., 2021, "Deciphering the &Art' in Modeling and Simulation of the Knee Joint: Variations in Model Development," ASME J. Biomech. Eng., **143**(6), p. 061002.
[30] Rooks, N. B., Schneider, M. T. Y., Erdemir, A., Halloran, J. P., Laz, P. J., Shelburne, K. B., Hume, D. R., et al., 2021, "A Method to Compare Heterogeneous Types of Bone and Cartilage Meshes," ASME J. Biomech. Eng., **143**(11), p. 111002.
[31] Andreassen, T. E., Laz, P. J., Erdemir, A., Besier, T. F., Halloran, J. P., Imhauser, C. W., Chokhandre, S., et al., 2023, "Deciphering the &Art' in Modeling and Simulation of the Knee Joint: Assessing Model Calibration Workflows and Outcomes," ASME J. Biomech. Eng., **145**(12), p. 121008.
[32] Erdemir, A., and KneeHub Project Team, 2018 "Reproducibility in Simulation-Based Prediction of Natural Knee Mechanics: Project Home," Stanford, CA, accessed Oct. 1, 2025, https://simtk.org/projects/kneehub
[33] Chokhandre, S., Schwartz, A., Klonowski, E., Landis, B., and Erdemir, A., 2023, "Open Knee(s): A Free and Open Source Library of Specimen-Specific Models and Related Digital Assets for Finite Element Analysis of the Knee Joint," Ann. Biomed. Eng., **51**(1), pp. 10–23.
[34] Petersen, W., and Zantop, T., 2007, "Anatomy of the Anterior Cruciate Ligament With Regard to Its Two Bundles," Clin. Orthop. Relat. Res., **454**, pp. 35–47.
[35] Anderson, C. J., Ziegler, C. G., Wijdicks, C. A., Engebretsen, L., and LaPrade, R. F., 2012, "Arthroscopically Pertinent Anatomy of the Anterolateral and Posteromedial Bundles of the Posterior Cruciate Ligament," J. Bone Jt. Surg. Am., **94**(21), pp. 1936–1945.
[36] Hosseini, A., Qi, W., Tsai, T.-Y., Liu, Y., Rubash, H., and Li, G., 2015, "In vivo Length Change Patterns of the Medial and Lateral Collateral Ligaments Along the Flexion Path of the Knee," Knee Surg. Sports Traumatol. Arthrosc., **23**(10), pp. 3055–3061.
[37] Mommersteeg, T. J., Kooloos, J. G., Blankevoort, L., Kauer, J. M., Huiskes, R., and Roeling, F. Q., 1995, "The Fibre Bundle Anatomy of Human Cruciate Ligaments," J. Anat., **187**(Pt. 2), pp. 461–471.
[38] Kia, M., Schafer, K., Lipman, J., Cross, M., Mayman, D., Pearle, A., Wickiewicz, T., and Imhauser, C., 2016, "A Multibody Knee Model Corroborates Subject-Specific Experimental Measurements of Low Ligament Forces and Kinematic Coupling During Passive Flexion," ASME J. Biomech. Eng., **138**(5), p. 051010.
[39] Banovetz, M. T., Familiari, F., Kennedy, N. I., Russo, R., Palco, M., Simonetta, R., DePhillipo, N. N., and LaPrade, R. F., 2023, "Anatomy of the Anterior Cruciate Ligament and the Common Autograft Specimens for Anterior Cruciate Ligament Reconstruction," Ann. Jt., **8**, p. 28.
[40] Peters, A. E., Akhtar, R., Comerford, E. J., and Bates, K. T., 2018, "Tissue Material Properties and Computational Modelling of the Human Tibiofemoral Joint: A Critical Review," PeerJ, **6**, p. e4298.
[41] Grood, E. S., and Suntay, W. J., 1983, "A Joint Coordinate System for the Clinical Description of Three-Dimensional Motions: Application to the Knee," ASME J. Biomech. Eng., **105**(2), pp. 136–144.
[42] Engebretsen, L., Wijdicks, C. A., Anderson, C. J., Westerhaus, B., and LaPrade, R. F., 2012, "Evaluation of a Simulated Pivot Shift Test: A Biomechanical Study," Knee Surg. Sports Traumatol. Arthrosc.: Off. J. ESSKA, **20**(4), pp. 698–702.
[43] Harner, C. D., Vogrin, T. M., and Woo, S. L.-Y., 2001, "Anatomy and Biomechanics of the Posterior Cruciate Ligament," *Posterior Cruciate Ligament Injuries: A Practical Guide to Management*, G. C. Fanelli, ed., Springer, New York, pp. 3–22.
[44] Li, G., DeFrate, L. E., Sun, H., and Gill, T. J., 2004, "In vivo Elongation of the Anterior Cruciate Ligament and Posterior Cruciate Ligament During Knee Flexion," Am. J. Sports Med., **32**(6), pp. 1415–1420.
[45] Pourmodheji, R., Chalmers, B. P., Debbi, E. M., Long, W. J., Wright, T. M., Westrich, G. H., Mayman, D. J., Imhauser, C. W., Sculco, P. K., and Kahlenberg,

Downloaded from http://asmedigitalcollection.asme.org/biomechanical/article-pdf/148/5/051004/7584244/bio-25-1200.pdf by University of Denver user on 08 March 2026

C. A., 2024, "Impact of Selective Posterior Cruciate Ligament Fiber Release on Femoral Rollback in Cruciate-Retaining Total Knee Arthroplasty: A Computational Study," J. Arthroplasty, **39**(8), pp. S347–S352.e2.
[46] Marom, N., Amirtharaj, M. J., Jahandar, H., Shamritsky, D. Z., Tao, M. A., Ouanezar, H., Nawabi, D. H., Wickiewicz, T. L., Imhauser, C. W., and Pearle, A. D., 2025, "Compressive Force and Valgus Torque Are the Predominant Applied Loads During the Pivot Shift Exam: An In vitro Study," Knee Surg. Sports Traumatol. Arthrosc., **33**(6), pp. 2086–2094.
[47] Imhauser, C. W., Kent, R. N., Boorman-Padgett, J., Thein, R., Wickiewicz, T. L., and Pearle, A. D., 2017, "New Parameters Describing How Knee Ligaments Carry Force In Situ Predict Interspecimen Variations in Laxity During Simulated Clinical Exams," J. Biomech., **64**, pp. 212–218.
[48] Hara, K., Mochizuki, T., Sekiya, I., Yamaguchi, K., Akita, K., and Muneta, T., 2009, "Anatomy of Normal Human Anterior Cruciate Ligament Attachments Evaluated by Divided Small Bundles," Am. J. Sports Med., **37**(12), pp. 2386–2391.
[49] Gupte, C. M., Smith, A., McDermott, I. D., Bull, A. M. J., Thomas, R. D., and Amis, A. A., 2002, "Meniscofemoral Ligaments Revisited: Anatomical Study, Age Correlation and Clinical Implications," J. Bone Jt. Surg. Br., **84-B**(6), pp. 846–851.
[50] LaPrade, R. F., Ly, T. V., Wentorf, F. A., and Engebretsen, L., 2003, "The Posterolateral Attachments of the Knee," Am. J. Sports Med., **31**(6), pp. 854–860.
[51] Liu, F., Yue, B., Gadikota, H. R., Kozanek, M., Liu, W., Gill, T. J., Rubash, H. E., and Li, G., 2010, "Morphology of the Medial Collateral Ligament of the Knee," J. Orthop. Surg., **5**(1), p. 69.
[52] LaPrade, R. F., Engebretsen, A. H., Ly, T. V., Johansen, S., Wentorf, F. A., and Engebretsen, L., 2007, "The Anatomy of the Medial Part of the Knee," J Bone Jt. Surg., **89**(9), pp. 2000–2010.
[53] Smith, C. R., Lenhart, R. L., Kaiser, J., Vignos, M. F., and Thelen, D. G., 2015, "Influence of Ligament Properties on Tibiofemoral Mechanics in Walking," J. Knee Surg., **29**(02), pp. 99–106.
[54] Rooks, N. B., Besier, T. F., and Schneider, M. T. Y., 2022, "A Parameter Sensitivity Analysis on Multiple Finite Element Knee Joint Models," Front. Bioeng. Biotechnol., **10**, p. 841882.
[55] Baldwin, M. A., Laz, P. J., Stowe, J. Q., and Rullkoetter, P. J., 2009, "Efficient Probabilistic Representation of Tibiofemoral Soft Tissue Constraint," Comput. Methods Biomech. Biomed. Eng., **12**(6), pp. 651–659.
[56] Naghibi Beidokhti, H., Janssen, D., van de Groes, S., Hazrati, J., Van den Boogaard, T., and Verdonschot, N., 2017, "The Influence of Ligament Modelling Strategies on the Predictive Capability of Finite Element Models of the Human Knee Joint," J. Biomech., **65**, pp. 1–11.
[57] Theodorakos, I., and Andersen, M. S., 2024, "Influence of the Initial Guess on the Estimation of Knee Ligament Parameters Via Optimization Procedures," Bioeng. Basel Switz., **11**(12), p. 1183.

Downloaded from http://asmedigitalcollection.asme.org/biomechanical/article-pdf/148/5/051004/7584244/bio-25-1200.pdf by University of Denver user on 08 March 2026